LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	17
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Tyre and tubes Tyre variation of Tyre pressure with all up weight, creep and creep marking on wheels
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Tyre and tubes Tyre variation of Tyre pressure with all up weight, creep and creep marking on wheels
5. Specific Objectives	17.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Inspection of tyre pressure variation with weight; identification of creep marks on wheels; safety precautions regarding hot tyres and jack placement; instructor supervision during ground handling simulation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aircraft tyres?
(b) How does aircraft weight affect tyre pressure requirements?
(c) What are the visible signs of tyre creep on a wheel?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Typical alighting gear construction • Tyre pressure variation with all up weight • Factors influencing pressure limits	L & Ds	PPT, IB	15 min
(b)	• Creep phenomenon explanation • Creep marking identification techniques • Safety protocols for creep assessment	L & Ds	PPT, PDF	20 min
(c)	• Practical inspection of tyre pressure • Identification of creep marks on wheels • Verification of pressure against weight limits	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how all-up weight influences the required tyre pressure.
(b) Describe the procedure for identifying creep marks on aircraft wheels.
(c) List three safety precautions when inspecting tyres after landing.

12. Home Assignment and Follow Up	Prepare a brief report on the relationship between aircraft all-up weight and tyre pressure, referencing the next lesson plan on Tyre and brake care.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Tyre and brake care	Refer to the master syllabus reference list	LP 18

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min